SUBHARUP BISWAS

Software & Security Engineer | Kolkata, India | +91 62899 25694
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TECHNICAL SKILLS

- **Languages & Web**: JavaScript, TypeScript, Python, SQL, HTML5/CSS3
- **Frameworks & DBs**: React, Next.js, Node.js, Express, Flask, Tailwind CSS, PostgreSQL, MongoDB, Prisma
- **Testing & Infrastructure**: Postman (API Testing), REST API Verification, Git, GitHub, Docker, Linux Admin
- **Security & Networking**: Wireshark, tcpdump, Snort, Ettercap, Vulnerability Scanning, JWT, Input Sanitization

TECHNICAL PROJECTS

PulsePing — AI-Driven Uptime & API Health Monitoring SaaS | *Next.js 16, React, TypeScript, Prisma, PostgreSQL*
github.com/SubharupBiswas/pulseping | pulseping.subharup.com

• Engineered a full-stack uptime SaaS using Next.js 16 and TypeScript; implemented automated health assertions and an anomaly diagnostic engine to classify root-cause API failures (35% faster render speeds).

VULNEXUS XDR — Full-Stack Extended Detection & Response Platform | *Python, Flask, SQLite, JWT, REST API*
github.com/SubharupBiswas/VULNEXUS

• Engineered a full-stack XDR engine featuring multithreaded port scanning, cross-platform OS security auditing, and automated CVE version matching.

• Built a malware inspection engine implementing Shannon entropy math models (H(X) > 7.5) and regex signatures alongside JWT-secured REST APIs.

SubnetMask.tech — IPv4 Matrix & Subnet Calculator | *TypeScript, React, Tailwind CSS, Bitwise Logic*
github.com/SubharupBiswas/cidr-subnet-calculator | subnetmask.tech

• Built a client-side network calculator executing low-latency IPv4 subnet calculations and real-time network mask visualizations via bitwise logic.

Rupee API Microservice — Real-Time Financial Data API | *Node.js, Express.js, Docker, REST APIs, Postman*
github.com/SubharupBiswas/RupeeCheck-AI | rupee.subharup.com

• Architected a containerized microservice delivering real-time currency exchange data; **validated REST API payloads, HTTP status codes, and edge-case handling using Postman.**

Self-Hosted Home Media & Storage Infrastructure | *Ubuntu Server, Samba, Jellyfin, Docker*
Environment: Private Local Network (Internal LAN / Self-Hosted Lab)

• Provisioned a headless Ubuntu Server running Samba file sharing and **Dockerized Jellyfin media services with 99.9% uptime and automated backups.**

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering (Cyber Security)
Techno Main Salt Lake, Kolkata, India — *7th Semester (Expected Graduation: May 2027)*

• **Core Coursework**: Software Engineering, Operating Systems, Computer Networks, DBMS, DSA, OOP

CERTIFICATIONS (CREDLY VERIFIED)

- **Programming & AI Badges**: Python Essentials 1 & 2 (Cisco) | Getting Started with Artificial Intelligence (IBM)
- **Cisco CCNA**: Introduction to Networks | Switching, Routing & Wireless | Enterprise Security & Automation